

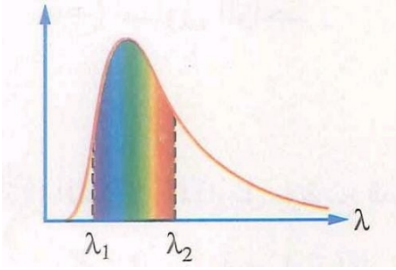
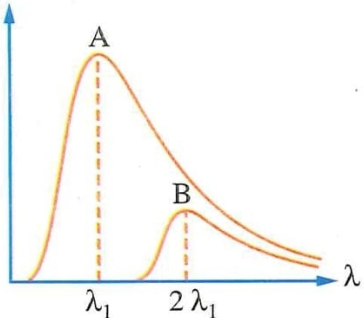
المعرفة	الكود
Radiation of Black Body and The Phenomenon	TPK01

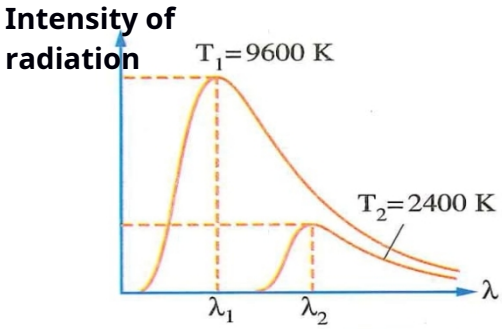
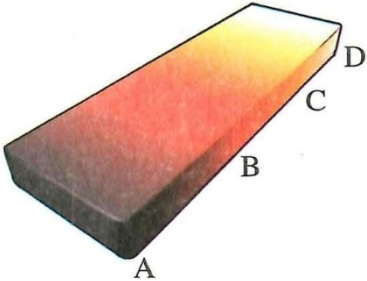
1- Choose the correct answer:

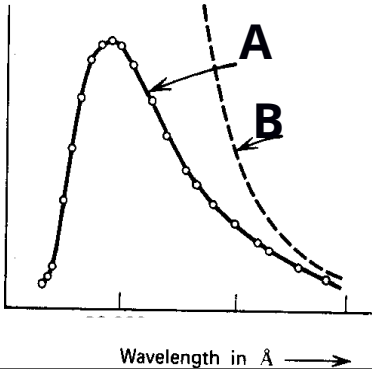
#	Question	Choices	Answer
1	Which of the following electromagnetic waves have the shortest wavelength	a. IR b. X-Rays c. UV d. Visible Light	b
2	The electromagnetic radiation that have the highest frequency and the shortest wavelength.....	a. Radio waves. b. Visible waves. c. Gama rays .	c
3	Due to the classical expectation the intensity of radiation increases by	a. Decreasing wavelength of radiation b. Decreases frequency of radiation c. Increases wavelength of radiation	a
4	the wavelength of a photon is 2.2×10^{-7} m, its energy is given by	a. 1×10^{35} joule b. 9×10^{-19} joule c. 3.3×10^{-27} joule d. 4.4×10^{-32} joule	$E = h \frac{c}{\lambda}$ $(6.62 \times 10^{-34}) \times (3 \times 10^8) / (2.2 \times 10^{-7})$ (b)
5	Difference between modern physics and quantum physics	a. Modern physics talks about macroscopical model b. Quantum physics talks about	c

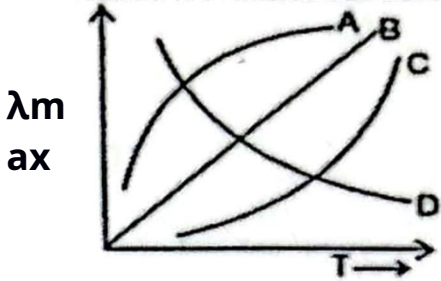
		macroscopical model c. Both are the same	
6	Difference between modern physics and classical physics	a. Modern physics talks about macroscopical model b. Classical physics talks about macroscopical model c. Both are the same	b
7	The emitted energy of the sun radiation is ----- visible light.	a. 40% b. 50% c. 20% d. 80%	a
8	The emitted energy of the sun radiation is ----- heat radiation.	a. 40% b. 50% c. 20% d. 80%	b
9	The emitted energy of the lamp radiation is ----- visible light.	a. 40% b. 50% c. 20% d. 80%	c
10	The emitted energy of the lamp radiation is ----- heat radiation.	a. 40% b. 50% c. 20% d. 80%	d
11	The electromagnetic radiation that have the lowest frequency and the tallest wavelength.....	a. Radio waves. b. Visible waves. c. Gama rays .	a
12	It explains all phenomena involved in electronics which is the basis for all modem electronic and	a. Classical physics. b. Quantum physics. c. All the above .	b

	communication systems.																	
1 3	The speed of electromagnetic waves equals	a. $3 \times 10^8 \frac{m}{s}$ b. $3 \times 10^{-8} \frac{m}{s}$ c. $4 \times 10^8 \frac{m}{s}$	a															
1 4	The speed of electromagnetic waves is calculated by the relation	a. $C = \cancel{v} \lambda \ v$ b. $C = \cancel{v} \lambda \ v$ c. $C = \cancel{v} \lambda - v$	a															
1 5	if the body is ideal black body , so its dominant color for the radiation that radiate from it is changes according to	a. the mass of the body. b. The density of the body. c. The temperature of the body. d. The type of the material that make the body.	c															
1 6	In the plank's curve for the black body radiation, if the temperature decreased	<table><tr><td></td><td>The peak of the curve</td><td>The space below the curve</td></tr><tr><td>A</td><td>Displaced to the high frequency region</td><td>Increase</td></tr><tr><td>B</td><td>Displaced to the high frequency region</td><td>Decrease</td></tr><tr><td>C</td><td>Displaced to the low frequency region</td><td>Increase</td></tr><tr><td>D</td><td>Displaced to the low frequency region</td><td>Decrease</td></tr></table>		The peak of the curve	The space below the curve	A	Displaced to the high frequency region	Increase	B	Displaced to the high frequency region	Decrease	C	Displaced to the low frequency region	Increase	D	Displaced to the low frequency region	Decrease	c
	The peak of the curve	The space below the curve																
A	Displaced to the high frequency region	Increase																
B	Displaced to the high frequency region	Decrease																
C	Displaced to the low frequency region	Increase																
D	Displaced to the low frequency region	Decrease																
1 7	In the Planck's curve for the black body radiation, the intensity of radiation at the very high frequencies is	a. Doesn't change. b. Moderate. c. Very high. d. Close to zero.	c															

1 8	At the opposite graph, if λ_1 is the shortest wavelength of the visible rays, while λ_2 is the longest wavelength for visible rays, so the Intensity of radiation  graph might represent the radiation that radiate from	a. A glowing star. b. Earth. c. Tungsten lamp. d. Human body.	a
1 9	The opposite graph represents the relation between the intensity of radiation and the wavelength for two hot bodies A,B, so the ratio Intensity of radiation  between their temperatures ($\frac{T_A}{T_B}$) is	a. $\frac{4}{1}$ b. $\frac{2}{1}$ c. $\frac{1}{2}$ d. $\frac{1}{4}$	b
2 0	The opposite graph shows a hot black body at T_1, T_2 so	a. $\frac{1}{16}$ b. $\frac{1}{8}$	c

	the ratio between $(\frac{\lambda_1}{\lambda_2})$ is Intensity of radiation 	c. $\frac{1}{4}$ d. $\frac{1}{2}$	
2 1	The opposite figure shows a hot iron piece, at which point that has the lowest temperature..... 	a. A b. B c. C d. D	a
2 2	A black body radiation with 3000 K, and its maximum wavelength λ, if the body is cooled to be T, its maximum wavelength become 10λ, so T equals	a. 300 K. b. 2700 K. c. 270 K. d. 1800 K.	a
2 3	In Planck's curve, the wavelength associating the maximum intensity of radiation emitted from the sun lies in region of	a. UV b. V.R c. I.R d. X-rays	b

2 4	The frequency of photons which emitted from a glowing body increases, then its number	a. Increases b. Decreases c. Remains constant d. Equals zero	b
2 5	Which of the following curves represents the classical explanation for blackbody radiation, and which one is the modern explanation 	a. A: modern B: classical. b. A: classical B: classical. c. A: modern B: modern. d. A: classical B: modern .	a
2 6	The distribution of the radiation intensity with wavelength is	a. Wein's distribution. b. Planck's distribution. c. Classical physics distribution. d. Modern physics distribution.	b
2 7	The wavelength at maximum radiation intensity is inversely proportional to the temperature is	a. Blackbody radiation. b. Wein's law. c. Planck's law.	b
2 8	For a black body radiation if $\lambda_{\max}$ represents the	a. A b. B	d

	maximum wavelength corresponding to maximum radiation at temperature T, the curve between λ_{max} and T is 	c. C d. D	
--	---	----------------------	--

2- Mention the factor affecting on

No.	Question	Answer
1	Energy of the photon	Frequency (directly) Wavelength (inversely)
2	The color of the hot objects	Temperature
3	The intensity of the black body radiation	Wavelength (inversely to a specific value then become directly) Frequency (directly to a specific value then become inversely)
4	The wavelength at maximum radiation intensity (λ_{max})	Temperature (inversely)
5	The speed of electromagnetic waves in air	It's constant (it is equal to the speed of light)
6	The speed of electromagnetic waves in media	The nature of the medium

3- True or False

No.	Question	Answer	Correction if F
1	when a piece of iron is heated to become self-luminous, the apparent color of the emitted radiation shifts initially from red to orange to yellow to blue and finally to white as the temperature increases	T	
2	The sun is yellow because its surface temperature is about 6000 K. Some stars glow blue-white because their surface temperature can be as high as 20000K	T	
3	Wave particle duality states that a wave can only act as a wave and not a particle.	F	The wave behave as a particle
4	Visible light is the part that can be seen with the human eye.	T	
5	Visible light is only a very large part of the electromagnetic spectrum.	F	Small part
6	Black body reflects no radiation and appears perfectly black.	T	
7	In Wien's law, the smaller temperature, the smaller the wave length of the peak	F	the higher temperature
8	All objects radiate energy continuously in the form of electromagnetic waves	T	

4- Mention the scientific term

No	Question	Answer
.		

1	It is the branch of physics that explains everything in our daily life and our common experiences	Classical physics
2	It is the branch of physics that deals with scientific phenomena in the universe which might not be directly observed in our daily life, and classical physics cannot explain	Modern (quantum) physics
3	They are vibrating electric and magnetic fields that moves through space or any other medium	Electromagnetic spectrum
4	The wavelength at maximum radiation intensity (λ_{max}) is inversely proportional to the temperature	Wien's Law
5	The wavelength (λ_{max}) at which the peak of the curve occurs is inversely proportional to temperature	Wien's Law
6	It is an ideal system that absorbs all radiation (100% of the radiation that hits it) falling on it. Then re-emit this radiation totally again	Black body radiation
7	The distribution of the radiation intensity with wavelength	Planck's distribution
9	A body that absorbs all the radiation falling on it with different wavelengths the emits them again in an ideal form .	Black body
10	The transfer of heat by electromagnetic waves such as visible light, infrared, and	Radiation

	ultraviolet radiation.	
11	the ratio between the photon energy to its frequency.	Planck's constant (h)
12	A phenomenon used for criminology and night vision	Thermal imaging

5- complete the following sentences:

No .	Question	Answer
1	is the transfer of heat by electromagnetic waves such as visible light, infrared, and ultraviolet radiation.	Radiation
2	 is the physics that explains everything in our daily life and our common experiences.	Classical physics
3	is resulting from vibrating electric and magnetic fields that moves through space or any other medium.	Electromagnetic waves
4	 is the part that can be seen with the human eye.	Visible rays
5	There are two types of body according to its radiation,,and	Glowing – non-glowing (opaque)
6	is that deals with scientific phenomena in the universe which might not be directly observed in	Modern (quantum) physics

	our daily life	
7	The wave particle duality means that the wave can behave as a ,and	Wave - particle
8	All objects radiate energy continuously in the form of produced by thermal vibrations of the molecules.	Electromagnetic waves
9	 deal with atomic and subatomic systems.	Modern (quantum) physics
10	All phenomena involved in electronics which is the basis for all modern electronic and communication systems is explained by	Modern (quantum) physics
11	its wavelength λ ranges from 400nm to 700nm.	Visible rays
12	Above 400nm is called, while below 700nm is called	Ultraviolet rays – infrared rays
13	According to $E=h\nu$, E is Proportional with ν .	Directly proportional
14	The atom release gained energy in the form of when it return into its original orbit .	Quanta
15	Radiation is composed of small particles called	Photons
16	40% of The sun radiation, and 20% of The lamp radiation is	Visible rays
17	50% of The sun radiation, and 80% of The lamp radiation is	Heat radiation

18	The maximum intensity of radiation increases by increasing According to Planck's Distribution.	Frequency
19	The wave length at maximum radiation intensity (λ_{max}) is..... proportional to the temperature.	Inversely
20	 is an ideal system that absorbs all radiation (100% of the radiation that hits it) falling on it. Then re-emit this radiation totally again	Black body
21	is a Prefect absorber and prefect emitter.	Black body
22	Classical explanation for black body radiation states that The intensity of radiation increases with the increasing of	Frequency
23	Major disagreement exists between classical theory and modern physics at	High frequency region (short wavelength)

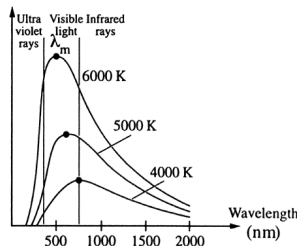
6- Solve the following:

No	Question	Answer
1	If the temperature of surface of sun is 6000 K and wavelength (λ_{max}) at maximum radiation intensity is 0.499 nm. Calculate the temperature of filament lamp that gives photons at wavelength (λ_{max}) = 1.2 nm	$\frac{\lambda_1}{\lambda_2} = \frac{T_2}{T_1} = \frac{0.499 \times 10^{-9}}{1.2 \times 10^{-9}} = \frac{T_2}{6000}$ $\frac{0.499 \times 10^{-9} \times 6000}{1.2 \times 10^{-9}} = 2495 K$

2	if the wavelength having a maximum intensity in the solar radiation is $0.5 \mu\text{m}$ Calculate the wavelength having a maximum intensity in the radiation emitted by Earth, given that the surface temperature of the sun and Earth are 6000 K and 300 K respectively.	$\frac{\lambda_1}{\lambda_2} = \frac{T_2}{T_1} = \frac{0.5 \times 10^{-6}}{\lambda_2} = \frac{300}{6000}$ Type equation here. $\frac{0.5 \times 10^{-6} \times 6000}{300} = 10 \times 10^{-5} \text{ m}$
3	calculate the wavelength of photon with energy = 4.968×10^{-19} Joule	$E = h\nu = h \frac{c}{\lambda}$ $\lambda = h \frac{c}{E}$ $\lambda = (6.62 \times 10^{-34}) \frac{(3 \times 10^8)}{(4.968 \times 10^{-19})}$ $\lambda = 4 \times 10^{-6} \text{ m}$
4	A body at temperature 1500 K radiates out maximum energy at the wavelength 20000 Å. If the sun radiates out maximum energy at 5000 Å, calculate the temperature of the sun	$\frac{\lambda_1}{\lambda_2} = \frac{T_2}{T_1}$ $\frac{20000}{5000} = \frac{T_2}{1500}$ $\frac{20000 \times 1500}{5000}$ = 6000K
5	What is the energy of a photon with frequency $3.0 \times 10^{14} \text{ Hz}$	$E = h\nu = (6.62 \times 10^{-34}) \times 3.0 \times 10^{14}$ = 1.99×10^{-19} joules
6	The wavelength of yellow light is approximately 5.9×10^{-7} meters. What is its frequency	$C = \lambda\nu$ $\nu = \frac{C}{\lambda} = \frac{3 \times 10^8}{5.9 \times 10^{-7}} = 5.1 \times 10^{14} \text{ Hz}$

7

- (a) What are the names of these curves?
- (b) Does the classical physics succeeded to explain these curves? And why?
- (c) What is meant by (λ_m) ?
- (d) What will happen to (λ_m) when the temperature decreases?



- (a) Planck's distribution
- (b) No , because at high frequencies the intensity of radiation decreases.
- (c) The maximum wavelength associating the maximum radiation intensity.
- (d) The maximum wavelength increases (inversely proportional)

7- Compare between the following:

1-

In terms of	Emitted rays from the sun	Emitted rays from the Earth
Region of maximum radiation	Visible ray	Infrared radiation

2-

In terms of	Emitted rays from the sun	Emitted rays from the electric lamp
-------------	---------------------------	-------------------------------------

Temperature	(T = 6000 K)	(T= 3000K)
The wavelength at the peak	$\lambda_{\text{max}} = 5000 \text{ \AA}$ (500 nm) (0.5 μm) (within the visible region)	$\lambda_{\text{max}} = 10000 \text{ \AA}$ (1000nm) (1 μm) (within the infra-red region)
the emitted energy	40% is a visible light 50% is a heat radiation 10% is a different region	20% visible light 80% heat radiation

3-

In terms of	Classical physics	Modern physics
Definition	It is that explains everything in our daily life and our common experiences.	It is that deals with scientific phenomena in the universe which might not be directly observed in our daily life.